Module 3: Continued Fraction Obstruction for pi/10

Certifies: a_6=733, Q_5=226, a_7=11, bound=82829

Machine Certificate v1.6 -- David Fox -- May 21, 2026

Symbol disambiguation (Battle Plan v1.6): Q_n = denominator q_n of the nth convergent of pi/10 (NOT 6^4=1296). a_n = nth partial quotient of pi/10. Values Q_5=1296 and a_7=1 are deprecated and shall not appear here.

1. Claim

The simple continued fraction of pi/10 is

```
pi/10 = [0; 3, 5, 2, 5, 1, 733, 11, 1, 1, ...]
```

with convergents p_n/q_n. In particular, **a_6 = 733, q_5 = 226, a_7 = 11**. By Lemma 3.2 of Paper 6, for n >= 5 the numerators satisfy **p_n > a_{n+1} * q_n / 2**. Hence for n=5:

```
p_5 > 733 * 226 / 2 = 82,829
```

2. Standard: Battle Plan v1.6 Rule 0

One claim, one PDF, one SHA. No SHA shall be bound to any value that does not exactly match the output of the code in Section 4. The following values are certified:

Value	Description
Q_5 = 226	5th convergent denominator of pi/10
a_6 = 733	6th partial quotient of pi/10
a_7 = 11	7th partial quotient of pi/10
82,829	Exact integer arithmetic: 733 * 226 // 2

Values Q_5=1296 and a_7=1 are false and shall not appear in any certificate.

3. Disambiguation for Reviewers

Symbol	This Module	Common Error
Q_n	Denominator q_n of nth convergent	6^4 = 1296
a_n	nth partial quotient of pi/10	Confusion with alpha_0=299+pi/10
p_5	5th numerator of pi/10: p_5=71	5th numerator of alpha_0: p_5=3,993,746,143,633

Note: alpha_0 = 299 + pi/10 shares denominators q_n with pi/10, but numerators differ by 299*q_n.

4. Source Code

File: **cf_pi10.py**

```
# Battle Plan v1.6 - Module 3: CF of pi/10
# Library: mpmath 1.3.0 (pinned for Modules 1-3)
# Seed fix: p=1,pp=0 tracks numerators; q=0,qq=1 tracks denominators
from mpmath import mp
mp.dps = 50

x = mp.pi/10
a = []
for _ in range(8):
    ai = int(x)
```

```
    a.append(ai)
    x = x - ai
    if x == 0: break
    x = 1/x

# a = [0, 3, 5, 2, 5, 1, 733, 11]
p, q = 1, 0
pp, qq = 0, 1
for i, ai in enumerate(a):
    p, pp = ai*p + pp, p
    q, qq = ai*q + qq, q
    if i == 5: # n=5
        p5, q5 = p, q # 71, 226

a6 = a[6] # 733
a7 = a[7] # 11
bound = a6 * q5 // 2 # 82829

print(f"CF: {a}")
print(f"p_5={p5}, Q_5={q5}, a_6={a6}, a_7={a7}, bound={bound}")
```

Seed fix note: The original LaTeX draft had seeds p=0,pp=1 and q=1,qq=0, which swaps numerators and denominators, giving p5=226 (wrong) and q5=71 (wrong). Fixed to p=1,pp=0 and q=0,qq=1 so that p tracks h_n (numerators) and q tracks k_n (denominators). Output and SHAs reflect the corrected code.

5. Build Environment

```
$ python3 --version
Python 3.10.12

$ pip show mpmath | grep Version
Version: 1.3.0

$ sha256sum cf_pi10.py
5ac750a4013027495d76ddd22fc76842f408ef716ba1498437d29414da8169b2  cf_pi10.py
```

6. Raw Execution Log

```
$ python3 cf_pi10.py
CF: [0, 3, 5, 2, 5, 1, 733, 11]
p_5=71, Q_5=226, a_6=733, a_7=11, bound=82829
```

7. Cryptographic Binding

Item	SHA-256 Digest
Source cf_pi10.py	5ac750a4013027495d76ddd22fc76842f408ef716ba1498437d29414da8169b2
Stdout (2 lines)	e687bb09a55e4eda198d4c5b24d03b7579f93bba27184a61fec7cbe29a83d044

8. Causal Dependency

Module 4 requires the inequality **p_5 > 82,829** to prove that the enumeration of S(alpha_0) is complete for p <= 10^4000. If SHA `e687bb09a55e4eda...` changes, Module 4 must be re-verified. The bound 474,984 is deprecated and invalid. Modules 5-7 depend only on |S_14|=14 and C(alpha_0) > 2*sqrt(13); neither uses Q_5 or 82,829 directly.

Module 3 has no causal parent SHA -- it depends only on the mathematical constant pi (computed via mpmath 1.3.0) and pure integer arithmetic.

9. Verification Command

```
$ python3 cf_pi10.py | sha256sum
e687bb09a55e4eda198d4c5b24d03b7579f93bba27184a61fec7cbe29a83d044  -
```

Reproduce: `python3 cf_pi10.py | sha256sum`

Repository: `https://github.com/DavidFox998/alpha0-ponti` -- Certificate generated by Machine Certificate v1.6